



Personal Directional Audio System (PDAS)

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Abstract

PDAS is a bone conduction headset for single sided hearing loss patients. Unlike traditional hearing aids, the PDAS provides a stylish bone conduction open ear headphone design. An analog microphone placed on the ear with the hearing loss starts the audio pipeline with the analog signal entering a microprocessor. PDAS contains an analog to digital converter; to achieve removal of noise and artefacts, it passes the digital signal conversion through a filter. Once the digital signal has been cleaned, PDAS then converts it back to analog, amplifies it, and transmits it through a bone oscillator sitting on the mastoid area of the normal hearing ear. The rerouted audio from the poor ear is transmitted to the normal ear through vibrations. The user has access to controls that allow them to turn off/on, as well as raise or lower the volume. PDAS is conveniently powered through a lithium-ion battery.

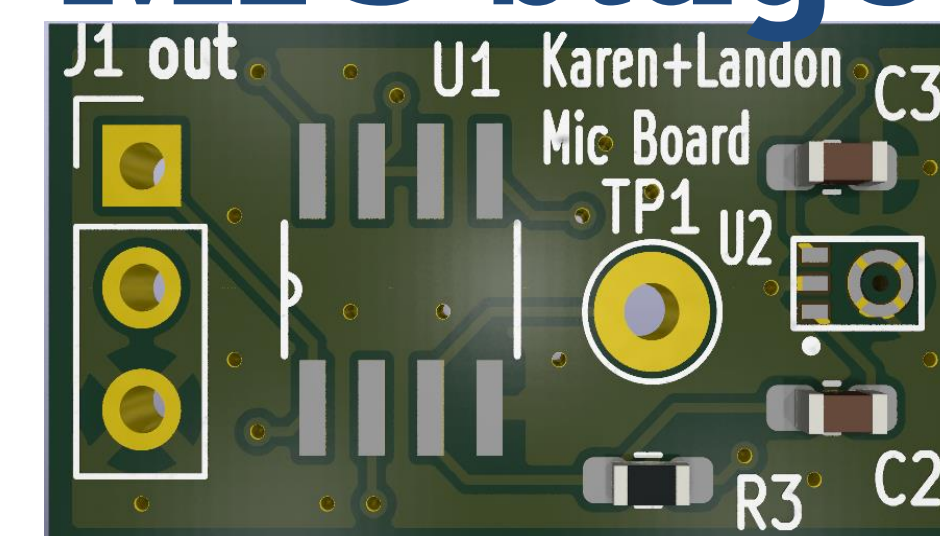
Project Objectives

- Design a PCB with hardware including mics, buttons, and an output speaker.
- Implement filtering to reduce noise heard by user by processing the sound in the processor (MCU).
- Validate performance by measuring the latency, delay, received by the microphone is not perceivable by the human ear (about 10ms).
- Design the headset to be rechargeable.

Conclusion

The PDAS is a bone conduction headset designed to help individuals with single-sided hearing loss by rerouting audio from the impaired ear to the functional ear. Using an analog microphone signal chain, the system achieves its core function with a delay of 20ms. Due to an amplifier failure during development, the final prototype outputs a noisy signal, though the fundamental audio rerouting performs as intended. Future iterations would have a smaller, lower-power MCU to reduce product size and improve wearability. The original design required multiple microphones to enhance performance in noisy environments. This goal that was scaled back due to size and battery constraints. A more compact, higher-capacity battery would also significantly improve the placement of the bone conduction.

MIC Stage



- Layout
- Analog mic (U2)
 - Amplifier Stage (U1)
 - Pins Out (J1 out)

Processing Stage

